

Limit Cycles of Model-Free PID Tuning by Step-Response Inspection

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Abstract

In a companion paper we introduced SPIN, a model-free method for tuning PID controllers: it draws three phase portraits from one closed-loop step response, counts a turn index in each, and changes the gains by a fixed table of four moves. The iteration does not stop at a point; it ends in a small periodic orbit. This paper explains that orbit and turns the explanation into an instrument.

In log-gain coordinates the four moves have exactly one convex combination summing to zero. This forces the orbit to circle the *triple point*, where all three turn indices sit on their thresholds, and fixes the move frequencies at $(\frac{1}{8}, \frac{1}{2}, \frac{1}{4}, \frac{1}{8})$; measured frequencies agree to three decimal places. The same computation over the integers gives the shortest possible orbit, of length eight, and bounds how far the gains stray from its centre. Stating this properly requires replacing the published integer count by an ε -regularized winding integral, used here only for the analysis.

Two practical results follow. The excursion bound fixes the step size from a required gain accuracy, with no plant parameter entering. And the balance condition read backwards measures the distance still to travel: the deviation of the observed move frequencies from the balance weights is the drift rate, to within a factor of 1.30 and again with plant-free constants. The rule can therefore read its own progress from the moves it has already made, with no extra experiment, and use that reading either to shrink its step or to stop tuning.

Index Terms— PID control, model-free tuning, limit cycles, switched systems, winding number

1 Introduction

Most PID tuning methods first fit a model and then compute the gains [2, 3]. In a companion paper [1] we took a different route and tuned directly from what the step response *looks like*. The method is called SPIN (Step-response Phase-portrait INspection). It draws three phase portraits from one closed-loop step response and counts how many times each curve turns around the origin. This count is the *turn index* N_k of band k , and it is

compared with a fixed threshold. A table of four moves, the *triangular rule*, then scales the gains up or down by a constant factor. No plant model is identified at any point.

The companion paper shows that the rule detunes aggressive gains, recovers from failed loops, and reaches a feasible region on a battery of standard plants. It also reports one behavior that it does not explain. No run stops at a point. Every successful run ends in a small periodic orbit in gain space. That orbit is the subject of this paper, and we ask four questions about it.

Where is it? Which point does the orbit circle, and is that point fixed by the plant or by the history of the run? *What is its shape and size?* Which moves occur, in what proportion, how short can an orbit be, and can the step size be chosen to meet a required accuracy on the gains? *Why this rule?* The moves are a design choice in the companion paper; is another choice possible? *Can the orbit be made smaller?* Is there a faster schedule whose accuracy the theory can guarantee rather than only observe?

One technical step comes first. The published turn index is an integer count, so it is discontinuous in the gains, and the boundaries between the regions of the rule are then not well defined. Section 3 replaces it by an ε -regularized winding integral. This is a tool for the analysis and not a change to the method: every sign comparison made by the published rule stays the same.

Our answers are then as follows. The orbit circles one point that depends only on the plant. We call it the *triple point*, because all three turn indices sit exactly on their thresholds there (Section 4). The move frequencies and the shortest possible length of eight both follow from a single balance condition on the four moves. The move table itself is forced: it is the only one compatible with the priority logic (Section 5). The same move counts bound how far the gains stray from the centre, which fixes the step size from a required accuracy (Section 6). Finally, the balance condition read backwards measures the distance still to be travelled, from the move sequence alone: the resulting drift meter tells the rule when to shrink its step and when to stop (Section 8).

Iterative model-free tuning is not new. Iterative feedback tuning [4] and extremum seeking [5] both estimate the gradient of a cost from closed-loop experiments and converge to a local minimum of it. The triangular rule is different: it uses only *signs*, with no cost function and no gradient. Its natural end state is therefore not a stationary point but a limit cycle of a switched system, which is

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exactly what averaged (Filippov) dynamics were developed for [7].

2 The Switched System

We recall only what is needed; the companion paper [1] defines the portraits, the index, and the rule in full. One closed-loop step experiment yields three curves (band 0, 1, 2), and for each curve a turn index $N_k \geq 0$ measuring how far the curve winds around the origin of its portrait. Each band has a fixed threshold:

$$\bar{N} = (\bar{N}_0, \bar{N}_1, \bar{N}_2) = (0.5, 0.75, 1.0). \quad (1)$$

The triangular rule reads the three comparisons $N_k \geq \bar{N}_k$ and applies one of four moves to the gain vector. Following [1] we order the gains by the band each channel dominates, $K^{(0)} = K_i$, $K^{(1)} = K_p$, $K^{(2)} = K_d$, so that N_k is the index pointing at $K^{(k)}$. If all three bands are below threshold, all gains are scaled up; otherwise the lowest offending band is cut by scaling gains down in a band-specific pattern. Every move multiplies each gain by $\gamma^{\pm 1}$ or leaves it alone, where $\gamma = 1/(1 - \beta)$ and β is the step size (default $\beta = 0.1$, so $\gamma \approx 1.111$). Multipliers are clipped to a box, here $[0.01, 10]$.

Throughout we assume the step response is measured without output noise, so that each turn index is a deterministic function of the gains. The stochastic case, in which the three comparisons become random near the target, is treated in follow-up work; the deterministic characterization below is its zero-noise limit, and Proposition 2 records a combinatorial property of the move table that bears on it.

We use log coordinates $\theta = (\log K_i, \log K_p, \log K_d)$. Note that these are in band order and not in the usual PID order. The four moves are then translations by $h = \log \gamma$:

$$\begin{aligned} e &= (+1, +1, +1) && \text{(expand),} \\ d_0 &= (-1, 0, 0) && \text{(cut band 0),} \\ d_1 &= (+1, -1, 0) && \text{(cut band 1),} \\ d_2 &= (+1, +1, -1) && \text{(cut band 2),} \end{aligned} \quad (2)$$

each scaled by h . The iteration is therefore a switched dynamical system. The signs of $N_k(\theta) - \bar{N}_k$ divide gain space into four regions, and each region is given one constant velocity. Since the velocity never vanishes inside a region, the state can never come to rest. A bounded trajectory must keep crossing from one region to another, so the rule must end in a cycle. Two questions remain: where the regions meet, and how the crossing pattern arranges itself there. Section 3 makes the first question well posed, and Section 4 answers both.

3 A Continuous Turn Index

Both questions concern the boundaries between regions, so before either can be answered we must check that

those boundaries exist. With the turn index as published they do not. That index is a hard count: the winding is accumulated along the portrait curve and then cut off when the curve enters a small disc of radius ε around the origin for the last time. The time of this *last entry* does not depend continuously on the curve. A very small change can make the curve just touch the edge of the disc, and one whole turn is then gained or lost. On the battery plants a gain change of 0.05%, far smaller than any step the rule takes, moves one index by exactly 1.000, and a longer simulation window does not help. The switching surfaces $N_k(\theta) = \bar{N}_k$ are then not surfaces at all, the four regions of Section 2 have no well-defined boundaries, and no statement about where they meet can even be formulated.

For running the rule this jump does no harm. The rule reads only the *sign* of $N_k - \bar{N}_k$, and a jump of one simply moves a switching surface a little. This is why all results of the companion paper remain valid. The analysis, however, needs a continuous index, and there is a natural one. For a curve $(x(t), y(t))$ we replace the hard cut-off by a smooth weight:

$$N_\varepsilon = \frac{1}{2\pi} \oint \frac{x dy - y dx}{x^2 + y^2 + \varepsilon^2}. \quad (3)$$

Write $\rho^2 = x^2 + y^2$. The integrand is then the usual winding integrand multiplied by the weight $\rho^2/(\rho^2 + \varepsilon^2)$. Turning far from the origin counts fully, turning at radius ε counts half, and turning deep inside the disc counts almost nothing. Three properties make N_ε the right tool.

It is continuous. The integrand is smooth everywhere, so N_ε is a continuous function of the gains. For the 0.05% change above it moves by 7×10^{-4} , where the hard count jumps by 1.000.

It does not depend on the window. After the response has settled the radius tends to zero, and there the weight tends to zero as ρ^2 . The value of N_ε is therefore exactly independent of the window length, and the δ -guard of the hard count is no longer needed.

It can replace the count. Away from the switching surfaces N_ε and the hard count agree up to an error that tends to zero with ε . Every sign comparison of the published rule is therefore unchanged, and so is every published result.

With N_ε the sets $N_k(\theta) = \bar{N}_k$ are true level surfaces, and the analysis of the next section is well posed. We also check directly that these surfaces meet in a single point. Runs in which the step size is driven to zero, which is a device for the analysis and not a tuning recommendation, converge to that point, which Section 4 calls θ^* , with final log-gain errors

$$\text{P1: } 0.010, \quad \text{P3: } 0.017, \quad \text{P4: } 0.041,$$

where θ^* is located independently by direct search on (5). With the hard count the same runs stop at points that depend on the schedule. This is further evidence

that only the regularized index reveals the true center of the cycle.

4 The Cycle and Its Center

We now use the regularized index of Section 3 throughout, so that the four regions have genuine boundaries. Suppose a trajectory stays bounded and does not drift. Over a long window, each move $v \in \{e, d_0, d_1, d_2\}$ is applied some fraction $w_v \geq 0$ of the time, with $\sum_v w_v = 1$. Non-drifting requires the average displacement $h \sum_v w_v v$ to vanish.

Proposition 1 (Unique Balance). *The only convex combination of the moves (2) that vanishes is*

$$\frac{1}{8} e + \frac{1}{2} d_0 + \frac{1}{4} d_1 + \frac{1}{8} d_2 = 0. \quad (4)$$

All four weights are strictly positive. This fixes the cycle in two ways.

Corollary 1 (Center of the Cycle). *A bounded, non-drifting trajectory must apply all four moves with the frequencies (4), and can therefore only orbit a point θ^* where all three switching surfaces meet:*

$$N_k(\theta^*) = \bar{N}_k, \quad k = 0, 1, 2. \quad (5)$$

We call θ^* the triple point. It depends only on the plant and the thresholds. It does not depend on the step size, on the starting gains, or on the history of the run.

The same computation over the integers gives a lower bound on the cycle length. No shorter cycle than the observed one can exist.

Corollary 2 (Minimal Period). *Any exactly closed orbit applies the moves an integer number of times $(n_e, n_{d_0}, n_{d_1}, n_{d_2})$ with $\sum_v n_v v = 0$, which forces $(n_e, n_{d_0}, n_{d_1}, n_{d_2}) = n_e \cdot (1, 4, 2, 1)$. Every closed orbit therefore has length a multiple of 8; the shortest possible cycle has length exactly 8, and no closed orbit can avoid any of the four moves.*

Table 1 tests these predictions on the whole battery and every plant confirms them exactly, the shares matching $(\frac{1}{8}, \frac{1}{2}, \frac{1}{4}, \frac{1}{8})$ to three decimal places in all four cases. Three plants close a period-8 orbit; P3 closes a period-16 orbit with counts $(2, 8, 4, 2)$. Corollary 2 predicts not merely period 8 but any multiple of eight with counts $n_e(1, 4, 2, 1)$, and P3 is the only member of the battery that tests the multiple.

Observation 1 (Ordering is Plant-Dependent). *The counts are universal but the order of the moves is not: each plant closes a different sequence. P2 gives the ruler ordering $d_0 d_1 d_0 d_2 d_0 d_1 d_0 e$, a cube of side h (Fig. 1); P4 gives $d_0 e d_0 d_2 d_0 d_1 d_0 d_1$ and P1 gives $d_1 d_1 e d_0 d_2 d_0 d_0 d_0$, flatter polygons of the same period and counts.*

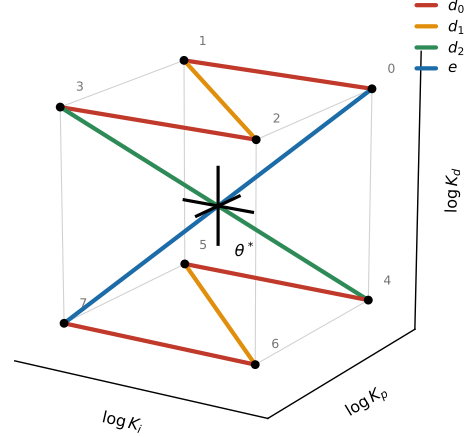


Figure 1: One period of the terminal orbit of P2, at $\beta = 0.10$ under the regularized index. Its measured ordering is the ruler sequence $d_0 d_1 d_0 d_2 d_0 d_1 d_0 e$, so the eight points are the corners of a cube of side h visited in binary counting order: $\log K_i$ is the least significant bit, $\log K_p$ and $\log K_d$ the higher bits, and e the carry that closes the period. The cross marks the centre. This is the tightest of the four orbits of Table 1.

Corollaries 1 and 2 fix the frequencies and the length; which ordering a plant selects is a property of its switching surfaces and remains a measurement. For practice this does not matter, because the counts alone bound the excursion whatever the ordering turns out to be.

Corollary 3 (Excursion Bound). *Let a closed orbit have counts $n_e(1, 4, 2, 1)$. Then over every ordering of those moves the coordinate span is at most $n_e(4, 2, 1)h$, so each coordinate of the orbit lies within*

$$n_e(2, 1, \frac{1}{2})h \quad (6)$$

of the orbit centre, in $(\log K_i, \log K_p, \log K_d)$.

Proof. Exhaustion over the orderings of $\{e, d_0^4, d_1^2, d_2\}$; the extreme spans are attained, so the bound is tight. The asymmetry reflects the table: K_i is moved by all four moves, K_d by two. $\square$

Table 1 reports the measured half-spans: every plant lies inside the bound, and P1 attains it in the proportional and derivative coordinates.

Boundary Case If θ^* lies outside the gain box the trajectory settles instead where, for each band, either the index sits on its threshold or the gain sits on a bound: a complementarity point. The companion paper's P2 run ends this way with K_d on the floor of 0.01, whereas from unit gains under the regularized index the same plant reaches an interior orbit (Table 1) — so this is a property of the starting gains, not of the plant.

Table 1: Terminal orbits of the whole battery, from 2000-iteration runs at $\beta = 0.10$ from unit gains with the regularized index (3). Every plant closes an *exactly* periodic orbit, no block departing from the pattern over the second half of the run, with move shares matching Proposition 1 to three decimals and counts $n_e(1, 4, 2, 1)$ as Corollary 2 requires. Half-spans are in units of h , in $(\log K_i, \log K_p, \log K_d)$. Under the published hard count the same runs miss the shares by up to 0.04 and close no orbit at all.

Plant	period	n_e	counts	half-span	bound
P1	8	1	(1, 4, 2, 1)	$(1.5, 1, \frac{1}{2})$	$(2, 1, \frac{1}{2})$
P2	8	1	(1, 4, 2, 1)	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$	$(2, 1, \frac{1}{2})$
P3	16	2	(2, 8, 4, 2)	$(3, 1.5, 1)$	$(4, 2, 1)$
P4	8	1	(1, 4, 2, 1)	$(\frac{1}{2}, 1, \frac{1}{2})$	$(2, 1, \frac{1}{2})$

5 The Move Table Is Forced

The results so far describe the cycle produced by one particular table of moves. It is natural to ask whether that table could have been chosen differently. It could not. Once the priority order is fixed, the table of Section 2 is the only one possible.

Proposition 2 (The Move Table is Forced). *Fix the priority order of the rule. Among move tables in which (i) every entry lies in $\{-1, 0, +1\}$, so that each gain is multiplied by $\gamma^{\pm 1}$ or left alone; (ii) $e = (1, \dots, 1)$; and (iii) each cut d_k decreases its own gain, $d_k[k] = -1$; the staircase table (2) is the unique one whose balance weights (4) coincide with the branch frequencies of the priority logic.*

Proof. Counting sign patterns, the priority logic selects e on 1 of the 2^n patterns and d_k on 2^{n-k-1} of them, so matching requires $e + \sum_k 2^{n-k-1} d_k = 0$. Coordinate j of this reads $\sum_k 2^{n-k-1} d_k[j] = -1$: a signed-binary representation of -1 with digits in $\{-1, 0, 1\}$. Exactly n such representations exist, indexed by the position m of the single -1 digit, namely $d_m[j] = -1$, $d_k[j] = +1$ for $k > m$ and 0 for $k < m$. Hypothesis (iii) forces $m = j$, giving $d_k[j] = +1$ for $j < k$, -1 for $j = k$, and 0 for $j > k$. This is exactly the table (2). $\square$

Hypotheses (i) and (iii) are both needed: without (iii) there are n^n matching tables, and relaxing (i) to entries in $\{-2, \dots, 2\}$ gives six for $n = 3$. The triangular shape is therefore not a design choice but the only shape that fits the selection logic together with the balance condition. The weights are powers of two because the selection tree is binary.

Two consequences follow. First, the matching gives a useful pairing: $\|d_k\| = \sqrt{k+1}$, while d_k occurs with frequency $2^{-(k+1)}$, so the most frequent move is always the smallest one. Second, the matching also says something about measurement noise. If noise removes all information from the three signs, the moves occur at the

branch frequencies, and those frequencies are the balance weights. The expected displacement is then zero, so the iteration stays at θ^* instead of drifting away. This is a property of the move table and is proved here by counting. A full stochastic treatment, including the case of correlated noise where this argument no longer holds, is left to future work.

6 Choosing the Step

The fixed-step iteration cannot converge to a point, and by Section 4 it should not be expected to. The practical question is how far from the target it leaves the gains, and Corollary 3 answers it without needing the ordering of the orbit: the excursion is bounded coordinate by coordinate, in units of the step. Halving β therefore halves it exactly.

Corollary 4 (Design Rule). *On a terminal orbit with $n_e = 1$, the gains stay within the multiplicative factors*

$$K_i : \gamma^2, \quad K_p : \gamma, \quad K_d : \gamma^{1/2}, \quad \gamma = (1 - \beta)^{-1}, \quad (7)$$

of their values at the orbit centre, at every iteration. To hold every gain within a common factor ρ it is therefore enough to choose

$$\boxed{\beta \leq 1 - \rho^{-1/2}} \quad (8)$$

The bound contains no plant parameter and is deliberately asymmetric: the integral gain is moved by all four moves and the derivative gain by two, so K_i is the loosest coordinate and K_d the tightest. At the companion paper's default $\beta = 0.1$ the guarantees are 23.5% on K_i , 11.1% on K_p and 5.4% on K_d ; holding all three within 5% requires $\beta \leq 0.024$. The multiplicity n_e is not assumed but read from the run, being the number of expand moves per period: it is 1 on three plants of the battery and 2 on P3 (Table 1), and the factors compound with it.

Three conditions come with (8). It is exact when the switching surfaces are locally flat, and curvature adds a term of order h^2 . It assumes that θ^* lies inside the gain box; if it does not, the boundary case of Section 4 applies. Under measurement noise it holds down to the noise floor, below which the spread around θ^* is set by the noise and not by β .

It is also worth saying which error is bounded. Equation (8) bounds the error in the *gains*, and this bound is the same for every plant. The corresponding error in the turn index is $|N_k - \bar{N}_k| \approx \|J_k\| \frac{\sqrt{3}}{2} h$, where J is the local Jacobian of Section 7, and this does depend on the plant. The indices are the measuring instrument; the gains are the result.

We measure this from far starting points, with the gains detuned by a factor of 16. With a constant β the run reaches the cycle quickly, and the final error of 0.074

is the radius of that cycle. This is a few percent in each gain, which for plant-floor use is indistinguishable from the target. This is the theoretical support for the companion paper’s constant-step recommendation: the fixed-step rule is guaranteed to leave the gains within one cycle radius of a plant-determined point, and that radius is set by β alone.

7 Local Structure

Two facts about the neighborhood of θ^* shape a proof. The central-difference Jacobians $\partial N_j / \partial \theta_i$ of the regularized index are nonsingular on every interior-target plant, with determinants 1.2, 5.9 and 4.3 on P1, P3, P4, so the center is locally unique. But they are neither diagonal nor diagonally dominant: on P1 the proportional gain moves the band-2 index harder than the derivative gain does, and on P4 the own-gain entry of band 1 is negative. The picture in which each move repairs its own band is therefore wrong at the center, and a per-move argument fails on two of three plants — yet on all three the cycle forms, centers correctly, and shrinks.

The resolution is that the cycle is a property of the *averaged* motion. Near θ^* the iteration switches rapidly between the four regions and its mean motion is the convex combination of their velocities weighted by residence time, which is the Filippov regularization of the switched system [7]; the measured shares of Table 1 are direct evidence, being exactly the balance weights (4). A proof should therefore show that θ^* is a stable equilibrium of the Filippov field and recover the discrete cycle by averaging. We leave this as the open problem. Note also that any scheme treating the bands separately, one step size or momentum term per band, assumes the diagonal structure that is missing here.

8 The Drift Meter

Everything so far describes where the cycle is and how big it is. What the running rule needs is different: it needs to know *whether it has got there yet*. This section shows that the balance condition answers that question too, and that the answer is already contained in the moves the rule has made, so no extra experiment is required.

The idea is a counting argument. Over a window of recent iterations, count how often each of the four moves was applied. If the shares match the balance weights w of (4) then by Proposition 1 the moves cancel: the gains have returned where they began and the state is circling the triple point. If they do not match, the gains are still travelling — and the *size* of the mismatch is not a flag but the speed of travel.

Proposition 3 (Drift Meter). *Let p be the vector of observed move shares over any window and M the move*

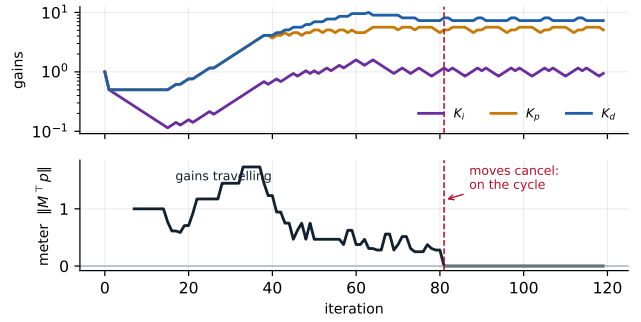


Figure 2: The meter on a run of the published rule (P1, $\beta = 0.1$, unit-gain start). Above, the gains; below, the meter over a sliding window of eight iterations. While the gains travel it reads of order one; at iteration 81 the moves begin to cancel and it falls identically to zero, the same instant the gains stop climbing and begin to cycle. Nothing but the move sequence enters the lower curve.

table (2). The mean displacement of the gains per iteration is

$$d = h M^T p = h M^T (p - w), \quad (9)$$

and its size is fixed by the mismatch $\|p - w\|$ within a factor of 1.30:

$$\sqrt{2} h \|p - w\| \leq \|d\| \leq 1.841 h \|p - w\|. \quad (10)$$

Proof. Each move v is applied a fraction p_v of the time and shifts θ by $h v$, which is the first equality. The second is $M^T w = 0$, that is (4). Since p and w are both share vectors, $p - w$ sums to zero; the two constants are the least and greatest singular values of M^T on that subspace, obtained from the move table alone. $\square$

A run climbing reads $\sqrt{3} h$ per iteration, one cutting only the lowest band reads $1 h$, one on the cycle reads zero (Fig. 2).

Three things make the meter practical. It contains no plant: the constants come from M , which the rule fixes, so they hold for every loop — unlike the local Jacobians of Section 7, whose condition numbers run into the hundreds. Its natural window is the period, since a closed orbit has counts $n_e(1, 4, 2, 1)$ by Corollary 2 and eight iterations therefore read identically zero. And it costs nothing, because the rule already records which row it fired.

Two uses *Shrink the step:* while the reading is large the state is travelling and the step should stay large; once it falls to zero the step buys no further progress and can be halved, handing the state to a concentric cycle of half the radius. *Stop tuning:* a zero reading also means nothing more can be learned by iterating, so the tuner may hold the gains and stop stepping the plant. Since the reading rises again as soon as the moves cease to cancel, the same statistic re-arms the tuner if the plant later

drifts, two thresholds giving the hysteresis that avoids chatter.

Why a fixed schedule will not do A step shrinking with the iteration number needs no detector, but does not work. From a $16\times$ detuned start the schedules $\beta_n \propto n^{-1/2}$ and $n^{-3/4}$ do reach θ^* , to 0.010 and 0.001, confirming the center; they are analysis instruments rather than recommendations. The classical n^{-1} stops early at error 0.197, for a reason that is a limit on total travel, not noise.

Lemma 1 (Logarithmic Travel). *Let $\beta_n = c/n$. The total distance covered in n steps is then $\sum_{m \leq n} \log \gamma_m = -\sum_{m \leq n} \log(1 - c/m) = O(\log n)$. A target at distance D therefore cannot be reached in fewer than about $n \approx e^{D/c}$ iterations, even if every step points the right way.*

With $D \approx \log 16$ per gain the required n is enormous — the same borderline case as in stochastic approximation, but arising here because the iteration is deterministic and the path is too short, not because of noise.

Application: a schedule that shrinks only when shrinking helps A schedule fixed in advance is either too slow or too weak as just shown. Proposition 3 removes the guesswork. While $\|p - w\|$ is large the state is traveling and the step should stay large; when it falls to the noise floor of the window the state is orbiting, the step is no longer buying progress, and it can be cut. In the limiting form of this test one halves β once every band has reversed, which is the cheapest sufficient condition for p to have visited all four moves. Each halving hands the state from one orbit to a concentric orbit of half the size, so the error falls geometrically at a bounded number of iterations per stage. On the interior-target plants this reaches 10^{-4} in 36 to 98 iterations, three orders of magnitude better than the constant step for a comparable number of experiments. This is a global relative of the sign-based Rprop update with step adaptation on reversal [6].

What the meter cannot see Under noise heavy enough to randomize the three comparisons, the priority logic fires at its branch frequencies, which by Proposition 2 are the balance weights themselves: the meter then reads zero although the state is not orbiting, reporting arrival when it has gone blind. The discriminating feature is the *ordering*, a genuine orbit being periodic at lag 8 while noise-driven moves share its marginals in independent order. A complete detector pairs (10) with a periodicity test, left to the stochastic treatment.

9 Conclusions

SPIN tuning with a fixed step ends in a limit cycle, and that cycle is now fully described. It circles the triple

point, the gain vector fixed by the plant at which all three turn indices sit on their thresholds. Its move frequencies $(\frac{1}{8}, \frac{1}{2}, \frac{1}{4}, \frac{1}{8})$ follow from a single balance condition, and the same condition over the integers gives the shortest possible period of 8. The cycle is realized as a binary counter on the orbit whose length is a multiple of eight with counts $n_e(1, 4, 2, 1)$, and those counts alone bound how far each gain strays from its centre. All four plants of the battery close such an orbit exactly, three with $n_e = 1$ and one with $n_e = 2$. If the triple point lies outside the gain box, the cycle instead settles at a complementarity point, as in the P2 run of the companion paper. Two tools made the analysis possible: the balance condition, which locates the center, and the ε -regularized winding index, which makes that center well defined.

The results support practice directly. The excursion bound keeps K_i , K_p and K_d within the factors γ^2 , γ , $\gamma^{1/2}$ of their values at the centre, so a common accuracy ρ follows from $\beta \leq 1 - \rho^{-1/2}$. This contains no plant parameter; at the companion paper's default $\beta = 0.1$ it guarantees 23.5%, 11.1% and 5.4% on the three gains.

The balance condition then closes the circle. Used forwards it locates the cycle; used backwards it measures the distance to it, since by Proposition 3 the deviation of the observed move frequencies from the balance weights is the drift rate, to within a factor of 1.30 and with constants that involve no plant. The rule can therefore read its own progress from the sequence of moves it has already made, needing no extra experiment, and shrink its step only when shrinking it helps: the resulting schedule reaches 10^{-4} accuracy in fewer than a hundred iterations. One theoretical step remains open, the existence and stability of the cycle through the averaged Filippov dynamics; and one practical limitation is known, that the meter cannot distinguish a true orbit from comparisons randomized by noise, for which the ordering of the moves rather than their frequencies is the discriminating statistic.

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